# Emma Dann

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## Education

**PhD in Biological Sciences (Computational Biology)** – University of Cambridge, UK (Oct 2019 – Oct 2023)
Advisors: Prof. Sarah A. Teichmann, Dr. John C. Marioni
Thesis title: *Discovering variation from cell atlases: comparative methods for single-cell genomics*

**MSc in Cancer, Stem Cells and Developmental Biology (Bioinformatics profile)** – Utrecht University, NL (Sep 2017 – Aug 2019)

**BSc in Biomolecular Sciences and Technology** – University of Trento, IT (Sep 2014 – July 2017)

## Academic appointments

**Post-doctoral fellow – Stanford University, Palo Alto, US** (May 2024 – present)
**Visiting post-doctoral fellow – Gladstone Institute for Genomic Immunology, San Francisco, US**
Hosted by Jonathan Pritchard and Alex Marson. Linking population genetic variation to immune traits with single-cell perturbation screens.

**Graduate researcher – Wellcome Sanger Institute, Cambridge, UK** (Oct 2019 – Jan 2024)
Supervised by Sarah Teichmann and John Marioni. Statistical methods for comparative analysis of single-cell omics data.

**Research intern – European Molecular Biology Laboratory, Heidelberg, DE** (Sep 2018 – Aug 2019)
Supervised by Wolfgang Huber. Analysis of multi-omic profiles of perturbations for functional characterization of drug treatments and target discovery.

**Research intern – KNAW Hubrecht Institute, Utrecht, NL** (Sep 2017 – Aug 2018)
Supervised by Alexander Van Oudenaarden. Biophysical modelling of whole-genome amplification via random priming in single-cell bisulfite sequencing experiments.

**Research intern – Center for Integrative Biology, University of Trento, IT** (Feb 2017 – July 2017)
Supervised by Francesca Demichelis. Epigenetic biomarker discovery from prostate cancer whole-methylome data.

## Publications

► denotes manuscripts as leading author, * denotes equal contribution

### Pre-print Manuscripts

► Zhu R.*, **Dann E.***, Yan J., Retana J.R., Goto R., … , Pritchard J.K., Marson A. (2025) Genome-scale perturb-seq in primary human CD4+ T cells maps context-specific regulators of T cell programs and human immune traits. *bioRxiv* | Full text

Deconinck L, Zappia L, Cannoodt R, Morgan M, **scverse core**, … , Saeys Y. (2025) *anndataR* improves interoperability between R and Python in single-cell transcriptomics. *bioRxiv* | Full text

Pett J.P., Prete M., Pham D., England N., … , **Dann E.**, … , Teichmann S.A., Meyer K. (2025) An atlas of TF driven gene programs across human cells. *bioRxiv* | Full text

► **Dann E.***, Teeple E.*, Elmentaite R., … , Rinaldis E., Teichmann S.A. (2024) Estimating the impact of single-cell RNA sequencing of human tissues on drug target validation., *medRxiv* | Full text | code

## Peer-reviewed Manuscripts

Heumos L., Ji Y., … , **Dann E.**, … , Theis F.J. (2025) Pertpy: an end-to-end framework for perturbation analysis. *Nature Methods* | Full text | python package

Ota M., Spence J.P., Zeng T., **Dann E.**, Milind N., Marson A., Pritchard J.K. (2025) Causal modeling of gene effects from regulators to programs to traits: integration of genetic associations and Perturb-seq. *Nature* | Full text

Luecken M., Gigante S., Burkhardt D., Cannoodt R., … , **Dann E.**, … , Saeys Y., Theis F.J., Krishnaswamy S. (2025) Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology* | Full text

Schuster V., **Dann E.**, Krogh A., Teichmann S.A. (2024) multiDGD: A versatile deep generative model for multi-omics data. *Nature Communications* | Full text | python package

Yayon N., Kedlian V., Boehme L., … , **Dann E**, … , Uhlmann V., Notarangelo L.D., Germain R.N., Radtke A.J., Marioni J.C., Taghon T., Teichmann S.A. (2024) A spatial human thymus cell atlas mapped to a continuous tissue axis. *Nature* | Full text

Oliver A.J., Ni H., … , **Dann E.**, …, Elmentaite R., Teichmann S.A. (2024) Single-cell integration reveals metaplasia in inflammatory gut diseases. *Nature* | Full text

Missarova A., **Dann E.**, … , Marioni J.C. (2023) Leveraging neighborhood representations of single-cell data to achieve sensitive DE testing with miloDE. *Genome Biology* | Full text | R package

Sumanaweera D., Suo C., Muraro D., **Dann E.**, … , Teichmann S.A. (2023) Gene-level alignment of single cell trajectories. *Nature Methods* | Full text | python package

Barnes J. L., He P., … , **Dann E.**, … , Teichmann S.A., Meyer K. B., Nikolic M. Z. (2022) Early human lung immune cell development and its role in epithelial cell fate. *Science Immunology* | Full text

► **Dann E.**, Cujba A.M., Oliver A., Meyer K., Teichmann S.A., Marioni J.C. (2023) Precise identification of cell states altered in disease with healthy single-cell references, *Nature Genetics* | Full text | code | python package

Botting R.A., Goh I., … , **Dann E.**, … , Teichmann S.A. , Haniffa M. (2023) Yolk sac cell atlas reveals multiorgan functions during human early development. *Science* | Full text

Suo C., Polanski K., **Dann E.**, … , Tuong Z.K., Clatworthy M., Teichmann S.A. (2023) Dandelion uses the single-cell adaptive immune receptor repertoire to explore lymphocyte developmental origins, *Nature Biotechnology* | Full text | python package

Heumos L., … **Single-cell Best Practices Consortium**, … , Theis F.J. (2023) Best practices for single-cell analysis across modalities. *Nature Review Genetics* | Full text

► Suo C.*, **Dann E.***, … , Haniffa M., Teichmann S.A. (2022) Mapping the developing human immune system across organs. *Science* | Full text | code | data

► Lalchand V.*, Ravuri A.*, **Dann E.***, … , Teichmann S.A., Lawrence N.D. (2022) Modelling Technical and Biological Effects in scRNA-seq data with Scalable GPLVMs. *Proceedings of the 17th Machine Learning in Computational Biology meeting* | Full text

He P., Lim K., Sun D., … , **Dann E.**, … , Meyer K.B., Rawlins E.L. (2022) A human fetal lung cell atlas uncovers proximal-distal gradients of differentiation and key regulators of epithelial fates, *Cell* | Full text

Kleshchevnikov V., Shmatko A., **Dann E.**, …, Stegle O., Bayraktar O.A. (2022) Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature Biotechnology* | Full text | python package

► **Dann E.**, Henderson N.C., Teichmann S.A., Morgan M.D., Marioni J.C. (2022) Differential abundance testing on single-cell data using k-nearest neighbor graphs, *Nature Biotechnology* | Full text | R package | python package

Jardine L., Webb S., … , **Dann E.**, … , Göttgens B., Roberts I., Teichmann S. and Haniffa M. (2021) Blood and immune development in human fetal bone marrow and Down syndrome, *Nature* | Full text

Elmentaite R., Kumasaka N., Roberts K., Fleming A., **Dann E.**, … , James K.R., Teichmann S.A. (2021), Cells of the human intestinal tract mapped across space and time, *Nature* | Full text

Stephenson E., Reynolds G., Botting R.A., Calero-Nieto F.J., Morgan M.D., Tuong Z.K., Bach K., Sungnak W., … , **Dann E.**, … , Duncan C.J.A, Smith K., Teichmann S.A., Clatworthy M.R., Marioni J.C., Gottgens B., Haniffa M. (2020) Single-cell multi-omics analysis of the immune response in COVID-19, *Nature Medicine* | Full text

Beltran H., Romanel A., … , **Dann E.**, … , Benelli M., and Demichelis F. (2020) Circulating Tumor DNA Profile Recognizes Transformation to Castration-Resistant Neuroendocrine Prostate Cancer, *Journal of Clinical Investigation* | Full text

# Fellowships & awards

**Tahoe Deep Dive Hackathon, first prize winner** (May 2025)
Awarded $20,000 for best project on prediction of clinical outcomes from large-scale drug perturbation datasets.

**Helen Hay Whitney fellowship** (Nov 2024 – present)
Postdoctoral fellowship, Awarded $229,500 over 3 years

**European Molecular Biology Organization (EMBO) long-term fellowship** (Jul 2024 - present)
Postdoctoral fellowship, awarded €130,800 over 2 years (switched to non-stipendiary fellowship since Apr 2025)

**Wellcome Sanger Institute PhD studentship**, Wellcome Trust, UK (Oct 2019 - Oct 2023)
PhD fellowship, Awarded ~ £100,000 plus tuition over 4 years

**U/Select honours programme** Utrecht University, NL (Jan 2018 - Oct 2019)
Master's honors program, including €2000 travel grant for international research internship

**Merit Award for excellent graduates** University of Trento, IT (Oct 2017)
Prize for excellent graduates, awarded €3000

# Grants

**Contributed to NIH R01 grant (2R01HG008140)** (2025)
Title: Integration of genetic association mapping and functional data to elucidate genetic mechanisms of disease
PIs: Jonathan K. Pritchard, Alex Marson
Reviewed as 1st percentile score

**Contributed to ERC Advanced Grant (ThyDESIGN)** (2022)
Title: Learning from the thymic human cell atlas for T cell engineering
PIs: Sarah A. Teichmann
Awarded €2.5M over 5 years

# Presentations

## Conference oral presentations

Apr 2025 - Chan Zuckerberg Initiative Cell Science Meeting, Chicago (US)
Nov 2022 - Machine Learning in Computational Biology conference, virtual
Jun 2022 - Single Cell Biology conference, Wellcome Genome Campus, Hinxton (UK)
May 2022 - The Biology of Genomes, Cold Spring Harbor Laboratory (US)
Oct 2021 - Chan Zuckerberg Initiative Single Cell Biology Meeting, virtual
Sep 2021 - Next Generation Genomics Meeting, virtual
Mar 2021 - CZI Seed Networks Computational biology meeting, virtual

## Invited research seminars

Oct 2025 - Ultima Genomics user group session @ American Society for Human Genetics, Boston (US)
Sep 2025 - scverse community meeting, virtual
Jul 2025 - AI + biomedicine seminar, Stanford (US)
Sep 2024 - BRAID Genentech, South San Francisco (US)
Sep 2024 - Human Genetics department, University of Chicago, Chicago (US)
Sep 2024 - SQLIFTS Lecture Series, Northwestern University, Chicago (US)
Mar 2024 - Human Technopole, Milan (IT)
Dec 2023 - Google DeepMind, London (UK)

Jul 2023 - Sanofi Precision Medicine & Computational biology, Cambridge (US)
Jun 2023 - Chan Zuckerberg Initiative, Redwood City (US)
Apr 2023 - Cambridge AI club for biomedicine, Milner Institute, Cambridge (UK)
Dec 2022 - UMC Utrecht single-cell genomics meeting, virtual
Sep 2022 - UK Conference of Bioinformatics and Computational Biology, virtual
Jul 2022 - Computational Health Center seminar, Helmholtz Munich (DE)
Apr 2022 - Maxwell Society Annual Conference, King's College London (UK)
Apr 2022 - NYU Langone single cell journal club, virtual
Mar 2021 - Cambridge Center for Physical Biology Single Cell Symposium, virtual

### Guest lectures

Aug 2024 - Why there's never been a better time to study human biology, Girls in Research program (virtual)
Aug 2023 - Intro to single-cell genomics series, University of Sri Jayewardenepura (virtual)
Mar 2023 - Introduction to multiomics data integration and visualisation, EMBL-EBI training course (UK)
Apr 2022 - Best practices and open challenges in single-cell genomics data analysis, Maxwell Society Annual Conference, King's College London (UK)

### Conference poster presentations

Oct 2025 - American Society for Human Genetics conference, Boston (US)
Jul 2023 - Human Cell Atlas General Meeting, Toronto (CA)
Oct 2022 - Single Cell Genomics conference, Utrecht (NL)
Jun 2021 - Human Cell Atlas General Meeting, virtual

## Teaching & mentoring

**Computational research assistant recruitment and mentoring, Marson lab** (Fall 2025 - present)

**PhD rotation student recruitment and mentoring, Pritchard lab** (Winter 2025 - present)

**Stanford & Tanzania Human Genetics Organization virtual training program** (Spring 2025)
One-on-one virtual mentoring program on research project for students from Tanzania

**PhD rotation student recruitment and mentoring, Teichmann lab** (Fall 2021 - Fall 2023)

**Systems biology: From large datasets to biological insight, EMBL-EBI training course** (Jul 2022)
Invited instructor for single-cell multi-omics module (course material)

**Advanced topics in Single Cell Genomics, Swiss Institute for Bioinformatics course** (Apr 2022)
Invited instructor for multi-omics module (course material)

**Advanced topics in Single Cell Genomics, SciLifeLab / Swiss Institute for Bioinformatics Summer School** (Aug 2021)
Invited instructor for multi-omics module (course material)

## Leadership & outreach

**Core team member, scverse consortium** (Feb 2024 – present)
I am part of the governance team of the consortium, overseeing the development and maintenance of foundational software tools for omics data analysis.

- Organizational governance and strategic planning
- Development of academic and industry partnerships for funding initiatives and collaborative research projects
- Lead organizer of the scverse annual conference and community hackathons
- Contributor to open-source tutorials and resources for newcomers to single-cell genomics analysis in python

**Competition judge, Open Problems Single-cell perturbations Kaggle competition** (Fall 2023)
Served as judge for NeurIPS-affiliated machine learning challenge focused on predicting drug perturbation responses.

**Co-organizer, Strengthening integration pipelines session, Human Cell Atlas general meeting** (July 2023)
Co-organized and led panel discussion on cell atlas construction and usage.

**Member of the selection panel, Sanger prize competition** (Mar 2023)
The Sanger Prize competition funds research internships at Sanger for undergraduate students from LMI countries. I was a member of the Grad Panel, evaluating essays from applicants.

**Co-organizer Open Problems in Single-Cell Analysis Jamboree** (April 2021)
Led task for benchmarking of compositional analysis methods for scRNA-seq.

**Co-organizer, EBI-Sanger-Cambridge PhD Symposium** (Feb 2020)
I was in charge of the scientific program and sponsorship for the PhD program symposium.

**Board member, Open Wet Lab (OWL)** (Oct 2015 – Jan 2018)
OWL is the first biohacking organization in Italy. I was in charge of the association's budget and directed the publication of OWL's blog and scientific outreach articles for local newspapers.

# Peer review

**Journals:** Nature (1 paper), Cell (1* paper), Nature Biotechnology (2* papers), Nature Genetics (1 paper), Bioinformatics Advances (1 paper), OUP Bioinformatics (1 paper)
*Assisting a senior reviewer

**Conferences:** Machine Learning in Computational Biology

**Grants/fellowships:** Chan Zuckerberg Initiative Single Cell Biology Data Insights, Christian Doppler Research Association Transfer.S2S grants

# Software

**pertpy** | scverse/pertpy | contributor | perturbation analysis toolkit for the scverse ecosystem.

**miloR** | MarioniLab/miloR | primary developer | R/Bioconductor package for the identification of differential cell abundance in KNN graphs for single-cell data

**milopy** | emdann/milopy | primary developer | python package for the identification of differential cell abundance in KNN graphs for single-cell data

**genomic-features** | scverse/genomic-features | primary developer | Genomic annotations using Bioconductor resources in Python.

**oor_benchmark** | MarioniLab/oor_benchmark | primary developer | python framework to benchmark workflows for detection of out-of-reference cells in comparative single-cell analysis

**cell2location** | BayraktarLab/cell2location | contributor | python package for mapping of tissue cell types via integration of single cell RNA-seq and spatial transcriptomics data

# Industry experience

**Data science consultant – Ensocell Ltd** (Oct 2023 – Mar 2024)
Target discovery platform and analysis of single-cell data